EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Ionic Equilibrium In Solutions, Arrhenius Concept Of Acids And Bases, The Bronsted-Lowry Acids And Bases, Lewis Acids And Bases

1. The conjugate acid of $O^{2\ominus}$ ions is:

(A) $O_2^{\oplus}$ (B) $H^{\oplus}$ (C) $H_3O^{\oplus}$ (D) $OH^{\ominus}$

CIE099

2. Which ion does not show acidic behaviour:

(A) $[Al(H_2O)_6]^{3\oplus}$ (B) $[Fe(H_2O)_6]^{3\oplus}$ (C) $HPO_4^{2\ominus}$ (D) $ClO_3^{\ominus}$

CIE100

3. In the reaction $NH_3 + H_2O \rightleftharpoons NH_4^{\oplus} + OH^{\ominus}$, water behave as :

(A) Acid (B) Base (C) Neutral (D) Both acid & Base

CIE101

4. The conjugate base of $(CH_3)_2\overset{\oplus}{N}H_2$ is :

(A) CH_3NH_2 (B) $(CH_3)_2N^{\oplus}$ (C) $(CH_3)_2N$ (D) $(CH_3)_2NH$

CIE102

5. Weakest Bronsted Lowry base is :

(A) $H^{\ominus}$ (B) $OH^{\ominus}$ (C) $Cl^{\ominus}$ (D) $HCO_3^{\ominus}$

CIE103

6. Conjugate base of hydrazoic acid is:

(A) HN_3 (B) $N_3^{\ominus}$ (C) $N^{3\ominus}$ (D) $N_2^{\ominus}$

CIE104

7. In the reaction $HNO_3 + H_2O \rightleftharpoons H_3O^{\oplus} + NO_3^{\ominus}$, the conjugate base of HNO_3 is :

(A) H_2O (B) $H_3O^{\oplus}$ (C) $NO_3^{\ominus}$ (D) $H_3O^{\oplus}$ and $NO_3^{\ominus}$

CIE105

8. For the reaction $NH_4^{\oplus} + S^{2\ominus} \rightleftharpoons NH_3 + HS^{\ominus}$, NH_3 and $S^{2\ominus}$ are a group of :

(A) Acids (B) Bases (C) Acid-base pair (D) None of these

CIE106

9. The conjugate acid of $NH_2^{\ominus}$ is

(A) NH_3 (B) NH_2OH (C) $NH_4^{\oplus}$ (D) N_2H_4

CIE107

10. Which of the following pair is Lewis acid & Lewis base & Product of these is also Lewis base

(A) BF_3, NH_3 (B) $SiCl_4$, $2Cl^{\ominus}$ (C) $CH_3^{\oplus}$, $^{\ominus}OC_2H_5$ (D) All of these

CIE108

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Electrolytes, Ionization Constant Of Water, Ionic Product Of Water (K_w), Ionization Of Acids And Bases

11. Ionic product of water will increase, if we:

(A) Decrease the pressure (B) Add $H^{\oplus}$

(C) Add $OH^{\ominus}$ (D) Increase the temperature

CIE109

12. At 25°C, the dissociation constant for pure water is given by:

(A) $(55.4\times10^{14})^{-1}$ (B) 1×10^{-14}

(C) $\frac{1\times10^{-14}}{18}$ (D) None of these

CIE110

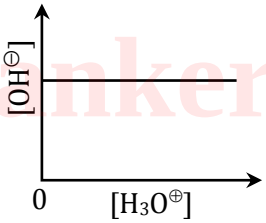
13. At 90°C, pure water has $[H_3O^{\oplus}] = 10^{-6.7}$ mol L^{-1} what is the value of K_w at 90°C:

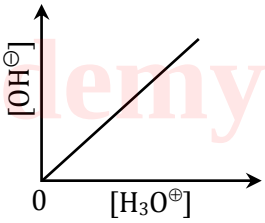
(A) 10^{-6} (B) 10^{-12}

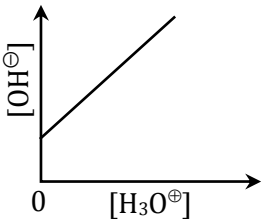
(C) 10^{-67} (D) $10^{-13.4}$

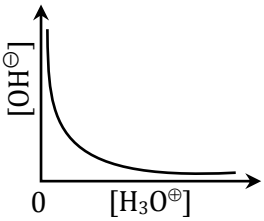
CIE113

14. Which of the following graphs describes the relationship between $[H_3O^{\oplus}]$ and $[OH^{\ominus}]$ in an aqueous solution at a constant temperature?

(A) 

(B) 

(C) 

(D) 

CIE115

15. Degree of dissociation of 0.1 N CH_3COOH is :- (Dissociation constant = 1×10^{-5})

(A) 10^{-5} (B) 10^{-4} (C) 10^{-3} (D) 10^{-2}

CIE123

16. The concentration of $[H^{\oplus}]$ and concentration of $[OH^{\ominus}]$ of a 0.1 M aqueous solution of 2% ionised weak acid is [ionic product of water $=1 \times 10^{-14}$]

(A) 0.02×10^{-3} M and 5×10^{-11} M (B) 1×10^{-3} M and 3×10^{-11} M

(C) 2×10^{-3} M and 5×10^{-12} M (D) 3×10^{-2} M and 4×10^{-13} M

CIE125

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The pH Scale, Calculation Of pH Of Solution Containing Strong Acid And Strong Base, Dilution Case

17. Choose the correct relation at T = 25°C

(A) $\frac{pH+pOH}{14}=7$ (B) pH + pOH = 14 (C) pOH = 14 + pH (D) pH = 14 + pOH

CIE111

18. The pH of H_2O at its boiling point :

(A) < 7 (B) > 7 (C) 7 (D) 0

CIE112

19. pH value of a solution, whose hydronium ion concentration is 6.2×10^{-9} mol/ℓ, is :-

(A) 6.21 (B) 7.21 (C) 7.75 (D) 8.21

CIE114

20. The pH of solution is increased from 3 to 6. Its $H^{\oplus}$ ion conc. will be :-

(A) Reduced to half (B) Doubled

(C) Reduced by 1000 times (D) Increased by 1000 times

CIE117

21. pOH of $[\frac{1}{200}]$ mol/m^3 H_2SO_4 (aq.) solution at 25°C is-

(A) 2 (B) 5 (C) 9 (D) 12

CIE118

22. What is the quantity of NaOH present in 250 cc of the solution, so that it gives a pH = 13 :-

(A) 10^{-13} g (B) 10^{-1} g (C) 1.0 g (D) 4.0 g

CIE126

23. Which one of the following has highest pH:-

(A) Distilled water (B) 1 M NH_3

(C) 1 M NaOH (D) Water saturated with chlorine

CIE127

24. 8 gm NaOH and 4.9 gm H_2SO_4 are present in one litre of the solution. What is its pH

(A) 1 (B) 13 (C) 12 (D) 2

CIE128

25. 10 ml of $\frac{M}{200}$ H_2SO_4 is mixed with 40 ml of $\frac{M}{200}$ H_2SO_4. The pH of the resulting solution is

(A) 1 (B) 2 (C) 2.3 (D) none of these

CIE134

26. Which of the following solution will have pH close to 1.0?

(A) 100 ml of M/100 HCl + 100 ml of M/10 NaOH

(B) 55 ml of M/10 HCl + 45 ml of M/10 NaOH

(C) 10 ml of M/10 HCl + 90 ml of M/10 NaOH

(D) 75 ml of M/5 HCl + 25 ml of M/5 NaOH

CIE135

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Common Ion Effect

27. If equal volume of 0.05 M ammonium hydroxide solution is dissolved in 0.001 M ammonium chloride solution. What will be the $OH^{\ominus}$ ion concentration of this solution :
$K_b(NH_4OH) = 1.8 \times 10^{-5}$
(A) 3.0×10^{-3} (B) 4.6×10^{-4} (C) 9.0×10^{-3} (D) 9.0×10^{-4}

CIE151

Ostwald Dilution Law

28. Ostwald's dilution law is not applicable for strong electrolytes because:
(A) Strong electrolytes are completely ionised
(B) Strong electrolytes are volatile
(C) Strong electrolytes are unstable
(D) Strong electrolytes often contain metal ions

CIE119

Calculation Of pH Of Solution Of Acid-Base Mixtures Containing Weak Acids And Weak Bases

29. Boric acid is a weak monoprotic acid. It ionizes in water as
$B(OH)_3 + H_2O \rightleftharpoons B(OH)_4^{\ominus} + H^{\oplus}$; $K_a = 5.9 \times 10^{-10}$
Calculate pH of 0.3 M boric acid is :
(A) 5.23 (B) 4.87 (C) 4.05 (D) 6.50

CIE183

30. The pH of 0.15 M solution of HOCl ($K_a = 9.6 \times 10^{-6}$) is:
(A) 4.42 (B) 2.92 (C) 3.42 (D) None

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CIE121

31. The molarity of nitrous acid at which its pH becomes 2. ($K_a = 4.5 \times 10^{-4}$):
(A) 0.3333 (B) 0.4444 (C) 0.6666 (D) 0.2222

CIE122

32. The pH of a 0.02 M ammonia solution which is 5% ionised will be :-
(A) 2 (B) 11 (C) 5 (D) 7

CIE124

33. Which of the following expression(s) for the degree of dissociation of weak monobasic acid in aqueous solution is/ are incorrect?
(A) $\sqrt{\frac{K_a}{c}}$ (B) $\frac{K_a}{K_a + [H^+]}$ (C) $\frac{[H^+]}{K_a + [H^+]}$ (D) $\frac{1}{1 + 10^{(pK_a - pH)}}$

CIE190

Salt And Types Of Salts, Hydrolysis Of Salts And The pH Of Their Solutions, Monovalent Salts

34. Hydroxyl ion concentration $[OH^{\ominus}]$ in the case of sodium acetate can be expressed as (where K_a is dissociation constant of CH_3COOH and C is the concentration of sodium acetate):
(A) $[OH^{\ominus}] = (CK_w.K_a)^{1/2}$ (B) $[OH^{\ominus}] = C.K_w\sqrt{K_a}$
(C) $[OH^{\ominus}] = \left(\frac{C.K_w}{K_a}\right)^{1/2}$ (D) $[OH^{\ominus}] = C.K_a.K_w$

CIE129

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35. The highest pH value is of :
(A) 0.1 M NaCl (B) 0.1 M NH_4Cl
(C) 0.1 M CH_3COONa (D) 0.1 M CH_3COONH_4

CIE130

36. What will be the pH of 1.0 M ammonium formate solution. If $K_a = 1 \times 10^{-4}$, $K_b = 1 \times 10^{-5}$:
(A) 6.5 (B) 7.5 (C) 8.0 (D) 9.0

CIE131

37. At 90° C, the pH of 0.1M NaCl aqueous solution is:
(A) < 7 (B) > 7 (C) 7 (D) 0.1

CIE132

38. Which salt will not undergo hydrolysis :
(A) KCl (B) Na_2SO_4 (C) NaCl (D) All

CIE133

39. What is the percentage hydrolysis of NaCN in N/80 solution when the dissociation constant for HCN is 2×10^{-9} and $K_w = 1.0 \times 10^{-14}$
(A) 2 (B) 5.26 (C) 8.2 (D) 9.6

CIE136

40. The compound whose 0.1 M solution is basic is
(A) Ammonium acetate (B) Ammonium chloride
(C) Ammonium sulphate (D) Sodium acetate

CIE137

41. The pK_a of a weak acid, HA, is 4.80. The pK_b of a weak base, BOH, is 4.78. The pH of an aqueous solution of the corresponding salt, BA, will be :
(A) 8.58 (B) 4.79 (C) 7.01 (D) 9.22

CIE138

Pankaj's Academy JEE

Buffer Solution, Designing Of Buffer Solution, pH Of Buffer

42. Henderson equation pH – pK_a = 1 will be applicable to an acidic buffer when :
(A) [Acid] = [Conjugate base] (B) [Acid] × 10 = [Conjugate base]
(C) [Acid] = [Conjugate base] × 10 (D) None of these

CIE146

43. When 0.02 moles of NaOH are added to a litre of buffer solution, its pH changes from 5.75 to 5.80. What is its buffer capacity :
(A) 0.4 (B) 0.05 (C) – 0.05 (D) 2.5

CIE147

44. Which of the following solutions does not act as buffer :–
(A) $H_3PO_4 + NaH_2PO_4$ (B) $NaHCO_3 + H_2CO_3$
(C) $NH_4Cl + HCl$ (D) $CH_3COOH + CH_3COONa$

CIE148

45. The pH of blood is maintained by CO_2 and H_2CO_3 in the body and chemical constituents of blood. This phenomenon is called :
(A) Colloidal (B) Buffer action (C) Acidity (D) Salt balance

CIE149

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46. The buffer solution play an important role in :

(A) Increasing the pH value (B) Decreasing the pH value

(C) Keeping the pH constant (D) Solution will be neutral

CIE150

47. Calculate the pH of a buffer prepared by mixing 600 cc of 0.6 M NH_3 and 400 cc of 0.5 M NH_4Cl. K_b for $NH_3 = 1.8 \times 10^{-5}$, (log 1.8 = 0.26)

(A) 11.3 (B) 9.0 (C) 9.52 (D) 5

CIE152

48. pK_b for NH_4OH at certain temperature is 4.74. The pH of basic buffer containing equimolar concentration of NH_4OH and NH_4Cl will be:-

(A) 7.74 (B) 4.74 (C) 2.37 (D) 9.26

CIE153

49. In a buffer solution the ratio of concentration of NH_4Cl and NH_4OH is 1 : 1. When it changes in 2 : 1, what will be the value of pH of buffer ?

(A) Increase (B) Decrease (C) No effect (D) None

CIE154

50. The total number of different kind of acidic buffers obtained during the titration of H_3PO_4 with NaOH are :

(A) 3 (B) 1 (C) 2 (D) 0

CIE155

Indicator

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51. The rapid change of pH near the stoichiometric point of an acid-base titration is the basis of indicator detection. pH of the solution is related to ratio of the concentrations of the conjugate acid (HIn) and base ($In^{\ominus}$) forms of the indicator by the expression :

(A) $\log\frac{[HIn]}{[In^{\ominus}]} = pK_{In} - pH$ (B) $\log\frac{[HIn]}{[In^{\ominus}]} = pH - pK_{In}$

(C) $\log\frac{[HIn]}{[In^{\ominus}]} = pH + pK_{In}$ (D) $\log\frac{[HIn]}{[In^{\ominus}]} = pK_{In} - pH$

CIE156

52. Calculate the pH range in which an acid indicator with K_{acid} (indicator) $= 1.0 \times 10^{-5}$ changes colour when the concentration of the indicator is 1×10^{-3} M.

(A) 5 ± 1 (B) 11 ± 1 (C) 3 ± 1 (D) 8 ± 1

CIE157

53. In what pH range will a 1×10^{-4} M solution of an indicator will K_b (indicator) $= 1 \times 10^{-11}$ change colour?

(A) 7.0 ± 1 (B) 3.0 ± 1 (C) 5.5 ± 1 (D) 11.0 ± 1

CIE158

54. Phenolphthalein is a :

(A) Strong acid (B) Strong base (C) Weak base (D) Weak acid

CIE160

For More Join: @IITJEE_Advanced

55. pH-range of Methyl red indicator is:
(A) 4·2–6·2 (B) 6·8–10·8 (C) 8 – 9·6 (D) 6·8 – 8·2

CIE161

pH Curve

56. A certain indicator (an organic dye) has $pK_a = 5$.
For which of the following titrations may it be suitable?
(A) Acetic acid against NaOH.
(B) Aniline hydrochloride against NaOH.
(C) Sodium acetate against HCl.
(D) Barium hydroxide against oxalic acid.

CIE200

57. Indicator which is used in the titration of CH_3COOH & NaOH:
(A) Methyl orange (B) Methyl red (C) Phenolphthalein (D) Litmus

CIE159

58. At what pH does an indicator change colour if the indicator is a weak acid with $K_{ind} = 4 \times 10^{-4}$.
For which one(s) of the following neutralizations would the indicator not be useful
(A) $NaOH + CH_3COOH$ (B) $HCl + NH_3$
(C) $HCl + NaOH$ (D) None of the above

CIE162

59. In the volumetric estimation of HCl, if we make use of phenolphthalein as an indicator, which base is unsuitable for the titration :-
(A) NaOH (B) RbOH (C) KOH (D) NH_4OH

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CIE163

60. Phenolphthalein does not act as an indicator for the titration between :-
(A) KOH and H_2SO_4 (B) NaOH and CH_3COOH
(C) Oxalic acid and $KMnO_4$ (D) $Ba(OH)_2$ and HCl

CIE164

61. For weak acid and strong base titration, the indicator used is :-
(A) Potassium di-chromate (B) Methyl orange
(C) Litmus (D) Phenolphthalein

CIE165

Solubility Equilibria Of Sparingly Soluble Salts, Solubility And Solubility Product Constant (K_{sp})

62. The values of K_{sp} for the slightly soluble salts MX and QX_2 are each equal to 4.0×10^{-18}. Which is the correct option.
(A) $S_1 > S_2$ (B) $S_1 < S_2$ (C) $S_1 = S_2$ (D) None of the above

CIE166

63. How many mole CuI ($K_{sp} = 5 \times 10^{-12}$) will dissolve in 1.0 L of 0.10 M NaI solution
(A) 6×10^{-11} (B) 5×10^{-11} (C) 8×10^{-11} (D) 7×10^{-11}

CIE168

For More Join: @IITJEE_Advanced

64. If the solubility of AgCl (formula mass=143) in water at 25°C is 1.43×10^{-4} gm/100 ml of solution then the value of K_{sp} will be :–

(A) 1×10^{-5} (B) 2×10^{-5} (C) 1×10^{-10} (D) 2×10^{-10}

CIE171

65. An aqueous solution contains 0.02 M-$FeCl_2$ and 0.05 M-$FeCl_3$. The solubility products are 8×10^{-16} for $Fe(OH)_2$ and 4×10^{-28} for $Fe(OH)_3$. Identify the correct option(s) among the following regarding the precipitation of metal hydroxides.

(A) At pH = 9.0, neither $Fe(OH)_2$ nor $Fe(OH)_3$ will precipitate.

(B) At pH = 6.0, neither $Fe(OH)_2$ nor $Fe(OH)_3$ will precipitate.

(C) If pH of the solution is in between 5.3 and 7.3, only $Fe(OH)_3$ will precipitate but not $Fe(OH)_2$.

(D) If pOH of the solution is in between 6.7 and 8.7, only $Fe(OH)_2$ will precipitate but not $Fe(OH)_3$.

CIE201

Simultaneous Solubility, Solubility Of A_2B, AB_2 Type Of Salt And Solubility Of Complex Salts New Question

66. If the solubility of $PbBr_2$ is 'S' g molecules per litre, considering 100% ionisation its solubility product is :

(A) $2S^3$ (B) $4S^2$ (C) $4S^3$ (D) $2S^4$

CIE167

67. A saturated solution of CaF_2 is found to contain 4×10^{-4} M fluoride ion. Calculate the K_{sp} of CaF_2. Neglect hydrolysis.

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(A) 3.2×10^{-11} (B) 32×10^{-11} (C) 6.4×10^{-11} (D) 64×10^{-12}

CIE169

68. The solubility of A_2X_3 is y mol dm^{-3}. Its solubility product is

(A) $6\ y^2$ (B) $64\ y^4$ (C) $36\ y^5$ (D) $108\ y^5$

CIE170

69. One litre of saturated solution of $CaCO_3$ is evaporated to dryness, 7.0 g of residue is left. The solubility product for $CaCO_3$ is:-

(A) 4.9×10^{-3} (B) 4.9×10^{-5} (C) 4.9×10^{-9} (D) 4.9×10^{-7}

CIE172

70. A_3B_2 is a sparingly soluble salt of molar mass M (g mol^{-1}) and solubility x g lit^{-1}. The ratio of the molar concentration of $B^{3\ominus}$ to the solubility product of the salt is

(A) $108\ \frac{x^5}{M^5}$ (B) $\frac{1}{108}\frac{M^4}{x^4}$ (C) $\frac{1}{54}\frac{M^4}{x^4}$ (D) None

CIE173

71. Solubility of Ag_2CrO_4 ($K_{sp} = 4 \times 10^{-13}$) in 0.1 M K_2CrO_4 solution will be :–

(A) 10^{-3} M (B) 10^{-6} M (C) 4×10^{-6} M (D) 5×10^{-7} M

CIE174

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72. At 30°C, In which of the one litre solution, the solubility of Ag_2CO_3 (solubility product = 8×10^{-12}) will be maximum :-

(A) 0.05 M Na_2CO_3 (B) Pure water

(C) 0.05 M $AgNO_3$ (D) 0.05 M NH_3

CIE175

73. For the reaction : $[Ag(CN)_2]^{\ominus}(aq) \rightleftharpoons Ag^{\oplus}(aq) + 2CN^{\ominus}(aq)$, K_C at 25ºC is 4×10^{-19}. Calculate $[Ag^{\oplus}]$ in solution which was originally 0.1 M in KCN and 0.03 M in $AgNO_3$.

(A) 1.5×10^{-17} M

(B) 7.5×10^{-18} M

(C) 4.6875×10^{-19} M

(D) Value cannot be determined due to mathematical complications.

CIE176

74. The solubility product values of AgCl and AgCNS are 1.8×10^{-10} and 1.6×10^{-11} respectively. When a solution is saturated with both solids, calculate the ratio $[Cl^{\ominus}]/[CNS^{\ominus}]$ and also $[Ag^{\oplus}]$ in the solution : Assume no reaction of cation/anion.

(A) 1.125, 4×10^{-6} M (B) 11.25, 1.4×10^{-5} M

(C) 1.25, 4×10^{-5} M (D) 1.25, 4×10^{-6} M

CIE177

75. 2M solution of Na_2CO_3 is boiled in a closed container with excess of CaF_2. Very little amount of $CaCO_3$ and NaF are formed. If the solubility product (K_{sp}) of $CaCO_3$ is x and molar solubility of CaF_2 is y, find the molar concentration of $F^{\ominus}$ in resulting solution after equilibrium is attained.

(A) $\sqrt{\frac{8y^3}{x}}$ (B) $\sqrt{\frac{y^3}{x}}$ (C) $\sqrt{8y^3}$ (D) $\sqrt{\frac{8y^2}{x}}$

CIE178

76. Na_3PO_4 which should be added in 10 L of 1.0×10^{-5} M - $BaCl_2$ solution without any precipitation of $Ba_3(PO_4)_2$ is [Ksp of $Ba_3(PO_4)_2$] = 4×10^{-23}

(A) 2×10^{-4} gm (B) 0.328 gm (C) 0.164 gm (D) 0.82 gm

CIE184

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Ostwald Dilution Law

1. Which of the following will occur if a 0.1 M solution of a weak acid is diluted to 0.01 M at constant temperature?

(A) $[H^+]$ will decrease.
(B) pH will increase.
(C) Percentage ionization will increase.
(D) K_a will increase.

CIE191

Calculation Of pH Of Solution Of Acid-Base Mixtures Containing Weak Acids And Weak Bases

2. Which of the following result(s) is/ are correct for the equilibrium state in a solution originally having 0.1 M-CH_3COOH and 0.1 M-HCl?
The value of Ka for $CH_3COOH = 1.8 \times 10^{-5}$.
(A) $[H^+] = 0.1$ M
(B) $[CH_3COO^-] = 1.8 \times 10^{-5}$ M
(C) Degree of dissociation of acetic acid $= 1.8 \times 10^{-4}$
(D) $[H^+]$ from water $= 10^{-13}$ M

CIE188

3. Which of the following statement(s) is/are correct?
(A) The pH of 10^{-8} M-HCl solution is 8.
(B) The conjugate base of $H_2PO_4^-$ is HPO_4^{2-}.
(C) Autoprotolysis constant of water increases with the increase in temperature.
(D) When a solution of a weak monoprotic acid is titrated against a strong base, at half neutralization point, pH = $\frac{pK_a}{2}$.

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CIE189

4. The relation $pK_a = 14 - pK_b$ is true for which of the following pair(s)?
(A) $MeNH_3^+$, CH_3NH_2
(B) CH_3COOH, CH_3COO^-
(C) H_3O^+, OH^-
(D) Na^+, NaOH

CIE192

Salt And Types Of Salts, Hydrolysis Of Salts And The pH Of Their Solutions, Monovalent Salts

5. The equilibrium constant for the ionization of $RNH_2(g)$ in water as

$RNH_2(g) + H_2O(l) \rightleftharpoons RNH_3^+(aq) + OH^-(aq)$

is 10^{-6} at 25° C. Which of the following is/ are correct?
(A) pH of the solution is 11.0 at partial pressure of $RNH_2(g)$ = 1 bar.
(B) Forward reaction is favoured by the addition of HCl(aq).
(C) Forward reaction is favoured by the addition of $H_2O(l)$.
(D) Forward reaction is favoured by the addition of $RNH_2(g)$.

CIE194

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Hydrolysis Of Polyvalent Salts

6. H_2CO_3 ionizes as follows.

$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$; $K_1 = 4.0 \times 10^{-6}$

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$; $K_2 = 5.0 \times 10^{-11}$

Which of the following information is correctly related to 0.5 M – Na_2CO_3 solution?

(A) The degree of hydrolysis of Na_2CO_3 is 0.02.

(B) The pH of solution is 2.0.

(C) The pOH of solution is 2.0.

(D) $[H_2CO_3] = 2.5 \times 10^{-9}$ M

CIE202

Amphiprotic Salt Hydrolysis

7. The amino acid glycine (NH_2CH_2COOH) is basic because of its $-NH_2$ group and acidic because of its –COOH group. By a process equivalent to base dissociation, glycine can acquire an additional proton to form $\overset{\oplus}{N}H_3CH_2COOH$. The resulting cation may be considered to be a diprotic acid, since one proton from the –COOH group and one from the $-\overset{\oplus}{N}H_3$ group may be lost. The pK_a values for these processes are 2.22 and 9.78, respectively. For a 0.01 M solution of neutral glycine (log 1.7 = 0.22, log 6 = 0.78),

(A) the pH is 6.0.

(B) percentage of the glycine in the cationic form is 0.017%.

(C) the pOH is 6.0.

(D) percentage of the glycine in the anionic form is 0.017%.

CIE203

Buffer Solution, Designing Of Buffer Solution, pH Of Buffer

Pankers Academy JEE

8. Which of the following may act as buffer in aqueous solution?

(A) $NH_4Cl + NH_4OH$ (B) $CH_3COOH + CH_3COONa$

(C) $CH_3COONa + NaCl$ (D) Borax + Boric acid

CIE196

9. A buffer solution can be prepared from a mixture of

(A) NH_4Cl and NaOH in 2 : 1 mole ratio.

(B) CH_3COONa and HCl in 1 : 1 mole ratio.

(C) CH_3COONa and HCl in 2 : 1 mole ratio.

(D) CH_3COONa and HCl in 1 : 2 mole ratio

CIE197

10. Which of the following statements is/are correct?

(A) A buffer solution may contain a weak acid and its conjugate base.

(B) A buffer solution shows little change in pH on the addition of a small amount of acid or base.

(C) A buffer solution can be prepared by mixing a solution of ammonium acetate and acetic acid.

(D) The addition of solid potassium cyanide to water increases the pH of water.

CIE198

pH Curve

11. A 2.5 g impure sample containing weak monoacidic base (Molecular weight = 45) is dissolved in 100 ml water and titrated with 0.5-M HCl at 25° C. When 1/5 th of the base was neutralized, the pH was found to be 9 and at equivalent point, the pH of solution is 4.5 (log 2 = 0.3).

(A) K_b of base is less than 10^{-6}.

(B) Concentration of salt at equivalent point is 0.25 M.

(C) Volume of HCl used at equivalent point is 100 ml.

(D) Mass percentage of base in given sample is 80%.

CIE199

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Calculation Of pH Of Solution Of Acid-Base Mixtures Containing Weak Acids And Weak Bases

Paragraph for Question No. 12 to 13

The dissociation constant of acetic acid is 2.0×10^{-5}. An aqueous solution of acetic acid is prepared by dissolving 0.7 millimole acetic acid in sufficient water to get 10 m^3 of solution at 25°C.

12. What is the pH of the solution? (log 2 = 0.3, log 7 = 0.85, log 17 = 1.23, $\sqrt{4.49} = 2.1$)

(A) 7.15 (B) 6.77 (C) 6.85 (D) 7.0

CIE204

13. What is the concentration of unionized acetic acid in the solution?

(A) 0 (B) 4.9×10^{-10} M (C) 7×10^{-8} M (D) 2.45×10^{-10} M

CIE205

Calculation Of pH Of Solution Containing Di- And Polybasic Acids And Di-And Polyacidic Bases

Paragraph for Question No. 14 to 15

A volume of 100 ml of 0.1 M-H_3PO_4 solution is titrated with 0.5 M-NaOH solution till the second equivalence point. Then 10 ml of 0.5 M-HCl solution is mixed in the resulting solution. The dissociation constants of H_3PO_4 are 10^{-3}, 10^{-8} and 10^{-13}.

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14. pH at the second equivalence point is

(A) 13.0 (B) 8.0 (C) 5.5 (D) 10.5

CIE209

15. pH of the solution after adding HCl is

(A) 8.0 (B) 1.0 (C) 6.0 (D) 10.5

CIE210

Amphiprotic Salt Hydrolysis

Paragraph for Question No. 16 to 17

Amino acid glycine (NH_2–CH_2–COOH) exists as a zwitter ion in aqueous solution. The K_a and K_b values of glycine are 1.6×10^{-10} ($pK_a = 9.8$) and 2.5×10^{-12} ($pK_b = 11.6$), respectively. The K_a and K_b values are for zwitter ion of amino acid with the following structure [$\overset{+}{N}H_3$ CH_2 – COO^-].

16. What is the value of K_b for –NH_2 group in glycine?

(A) 4×10^{-3} (B) 1.6×10^{-10} (C) 6.25×10^{-5} (D) 2.5×10^{-12}

CIE218

17. An aqueous solution of glycine has pH

(A) nearly 7 (B) nearly 7.9 (C) nearly 6.1 (D) nearly 11.5

CIE219

For More Join: @IITJEE_Advanced

Buffer Solution, Designing Of Buffer Solution, pH Of Buffer

Paragraph for Question No. 18 to 20

Human blood has a narrow pH range of 7.3 – 7.4, which must be maintained for metabolic processes to function properly. To keep the pH in this range, it requires a delicate balance between the concentrations of the conjugate acid–base pairs, making up to the buffer system. The main buffer is a carbonic acid-hydrogen carbonate system, which involves the following equilibria.

$CO_2(g) \rightleftharpoons CO_2(aq.)$

$CO_2(aq.) + H_2O(l) \rightleftharpoons H_2CO_3(aq.)$

$H_2CO_3(aq.) + H_2O(l) \rightleftharpoons HCO_3^-(aq.) + H_3O^+(aq.)$

At the body temperature, the pK_a for carbonic acid is 6.40. However, the normal concentration of $CO_2(g)$ in the lungs maintains a ratio of $HCO_3^-(aq.)$ and $H_2CO_3(aq.)$ in blood plasma at about 8 : 1. The H_2CO_3 concentration in blood is largely controlled by breathing and respiration and HCO_3^- concentration is largely controlled by excretion in urine. If blood pH rises above 7.4, then a potentially life-threatening condition called alkalosis can result. It normally occurs in climbers suffering from oxygen deficiency at high altitude or in patients who are hyperventilating from severe anxiety (log 2 = 0.3).

18. The pH of blood at body temperature is

(A) 6.4 (B) 7.4 (C) 7.3 (D) 7.2

CIE212

19. The maximum permissible value of $\frac{H_2CO_3}{HCO_3^-}$ in human blood to just prevent alkalosis is

(A) 0.1 (B) 10 (C) 8.0 (D) 0.125

CIE213

20. Which of the following statement is correct?

(A) One way to treat alkalosis can be to get the patient to breathe more quickly so that the amount of CO_2 exhaled increases and the blood becomes deficient in CO_2.

(B) One way to treat alkalosis is to get the patient breathe into a bag so that the exhaled CO_2 is re-inhaled.

(C) The pH of blood is independent of the concentration of CO_2.

(D) Alkalosis cannot be controlled by breathing and respiration.

CIE214

Simultaneous Solubility, Solubility Of A_2B, AB_2 Type Of Salt And Solubility Of Complex Salts

Paragraph for Question No. 21 to 23

The solubility product of AgCN is 1.0×10^{-16} and the formation constant of $Ag(CN)_2^-$ is 1.5×10^{17}.

21. The solubility of AgCN in 0.02 M-KCN solution, assuming no complex formation, is

(A) 1.0×10^{-8} M (B) 5.0×10^{-15} M (C) 0.02 M (D) 5.0×10^{-14} M

CIE215

For More Join: @IITJEE_Advanced

22. The solubility of AgCN in 0.02 M-KCN solution, assuming complex formation, is

(A) 0.3 M (B) 0.02 M (C) 1.33×10^{-19} M (D) 1.87×10^{-2} M

CIE216

23. At what $[CN^-]$ in the solution, the solubility of AgCN will be minimum?

(A) 3.65×10^{-9} M (B) 1.33×10^{-17} M (C) 2.58×10^{-9} M (D) 5.48×10^{-8} M

CIE217

Paragraph for Question No. 24 to 27

Phosphoric acid is of great importance in fertilizer production. Besides, phosphoric acid and its various salts have a number of applications in metal treatment, food, detergent and toothpaste industries.

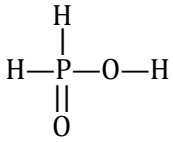
$pK_{a_1} = 2.12,\ pK_{a_2} = 7.21,\ pK_{a_3} = 12.32,$

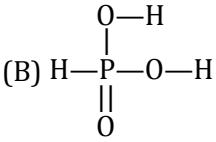
Small quantities of phosphoric acid are extensively used to impart the sour or tart taste to many soft drinks such as colas and root beers, in which a density of 1.00 g $mL^{\ominus 1}$ contains 0.05% by weight of phosphoric acid.

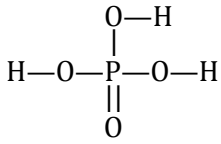
Phosphoric acid is used as a fertiliser for agriculture and in aqueous soil digesting. 1.00×10^{-3} M phosphoric acid is found to have pH = 7. Zinc is an essential micronutrient for plant growth. Plants can absorb zinc in water soluble from only. In the given soil, zinc phosphate is only the source of zinc and phosphate ions, K_{sp} (zinc phosphate) $= 9.1 \times 10^{-33}$.

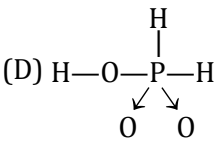
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24. Phosphoric acid is a tribasic acid with three-step ionisation constants. Thus, its structure is :

(A)


(B)


(C)


(D)


CIE179

25. What is the pH of the cola assuming that the acidity of the cola arises only from phosphoric acid and second and third ionisation constants are of no importance?

(A) 2.2 (B) 3.3 (C) 4.4 (D) 1.8

CIE180

26. Molar concentration of phosphate ion in the soil with pH 7 is :

(A) 1.2×10^{-4} M (B) 2.2×10^{-4} M (C) 1×10^{-3} M (D) 1.1×10^{-10} M

CIE181

27. Concentration of $[Zn^{2\oplus}]$ in the soil is :

(A) 9.1×10^{-5} M (B) 5.7×10^{-9} M (C) 4.0×10^{-10} M (D) 3.0×10^{-6} M

CIE182

For More Join: @IITJEE_Advanced

MATRIX MATCH TYPE QUESTION

Salt And Types Of Salts, Hydrolysis Of Salts And The pH Of Their Solutions, Monovalent Salts

28. Match the columns.

Column-I		Column-II	
(A)	NaCl	(P)	Cationic hydrolysis
(B)	Na_2CO_3	(Q)	Anionic hydrolysis
(C)	NH_4Cl	(R)	pH = 7.0 at 25 °C
(D)	CH_3COONH_4	(S)	pH < 7.0 at 25 °C
		(T)	pH > 7.0 at 25 °C

(A) A → P; B → Q, R; C → S, T; D → P, Q, S (B) A → R; B → Q, T; C → P, S; D → P, Q, R
(C) A → S; B → P, Q; C → S, Q; D → P, Q, T (D) A → R; B → S, T; C → Q, S; D → P, Q, S

CIE221

Amphiprotic Salt Hydrolysis

29. Match the columns.

Column-I		Column-II	
(A)	$\frac{pK_{a,H_2O} + pK_{b,H_2O}}{2}$ at 25°C (d_{water} = 1.0 g/ml)	(P)	$\frac{pK_w}{2}$
(B)	pH of CH_3COONH_4(aq) ($K_{a,CH_3COOH} = K_{b,NH_4OH}$)	(Q)	16 – log 1.8
(C)	pH of pure water at 320 K.	(R)	$-\log K_{a,H_2O}$
		(S)	< 7.0

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(A) A → P, R; B → Q; C → P, S (B) A → Q, R; B → P; C → P, S
(C) A → R, S; B → Q; C → R, S (D) A → Q, R; B → S; C → P, Q

CIE223

Buffer Solution, Designing Of Buffer Solution, pH Of Buffer

30. For a tribasic acid, H_3A, $K_{a1} = 10^{-4}$, $K_{a2} = 10^{-8}$ and $K_{a3} = 10^{-12}$. Match the pH (Column-II) of the resulting solution (Column-I) at 25 °C.

Column-I		Column-II	
(A)	Equimolar mixture of H_3A and NaH_2A.	(P)	12.0
(B)	Equimolar mixture of NaH_2A and Na_2HA.	(Q)	8.0
(C)	Equimolar mixture of Na_2HA and Na_3A.	(R)	4.0
(D)	Equimolar mixture of H_3A and NaOH.	(S)	6.0
(E)	Equimolar mixture of NaH_2A and NaOH.	(T)	10.0

(A) A → R; B → Q; C → P; D → S; E → T
(B) A → P; B → Q; C → R; D → S; E → T
(C) A → Q; B → S; C → R; D → P; E → T
(D) A → R; B → S; C → Q; D → P; E → T

CIE222

For More Join: @IITJEE_Advanced

31. Match the columns.

Column-I (pH value)		**Column-II (Solution)**	
(A)	5.0	(P)	pH at which $Mg(OH)_2$ just start precipitating from 2.0×10^{-3} M-Mg^{2+} solution (K_{sp} of $Mg(OH)_2 = 2 \times 10^{-6}\ M^3$).
(B)	9.0	(Q)	Maximum pH at which 0.1 mole of $Al(OH)_3$ dissolves completely as Al^{3+} in 1 L solution (K_{sp} of $Al(OH)_3 = 1 \times 10^{-28}\ M^4$).
(C)	12.5	(R)	pH at which 0.1-M-CH_3COOH dissociates 1000/11% (K_a of CH_3COOH-= 1.0×10^{-5} M).
(D)	6.0	(S)	pH of 0.001 M-HA (weak acid) solution (pK_a of HA = pK_b of A^-).
		(T)	pH of 3×10^{-5} M-CaA_2 solution (CaA_2 dissociates completely in water; K_a of HA = 5×10^{-9}).

(A) A → P, S; B → T; C → S; D → R
(B) A → Q, S; B → R; C → P; D → R
(C) A → Q, S; B → T; C → P; D → R
(D) A → P, S; B → R; C → S; D → R

CIE220

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

The pH Scale, Calculation Of pH Of Solution Containing Strong Acid And Strong Base, Dilution Case

1. Calculate the number of $H^{\oplus}$ present in one ml of solution whose pH is 13. If the number of $H^{\oplus}$ ions is $x \times 10^7$ then the value of x is

CIE021

2. The pH of the solution produced when an aqueous solution of strong acid ($p^H = 5$) is mixed with equal volume of an aqueous solution of strong acid of ($p^H = 4$) is :-

CIE022

3. Calculate pH of 10^{-7} M of NaOH solution

CIE030

Salt And Types Of Salts, Hydrolysis Of Salts And The pH Of Their Solutions, Monovalent Salts

4. How many grams of NH_4Cl should be dissolved per litre of solution to have a pH of 5.13? (K_b for NH_3 is 1.8×10^{-5}).

CIE025

Rankers Academy JEE

Hydrolysis Of Polyvalent Salts

5. The acid ionization (hydrolysis) constant of $Zn^{2\oplus}$ is 1.0×10^{-9} then the pH of a 0.001 M solution of $ZnCl_2$ is

CIE143

6. The extent of hydrolysis of 0.005 M K_2CrO_4 is x%, the value of 100x is the first ionization is essentially strong. [$K_2 = 3.2 \times 10^{-7}$ for H_2CrO_4]

CIE139

7. A 0.010 M solution of $PuO_2(NO_3)_2$ was found to have a pH of 4.0. The hydrolysis constant, K_h, for $PuO_2^{2\oplus}$ is 10^{-a} and K_b for $PuO_2OH^{\oplus}$ is 10^{-b}, then the value of a × b is

CIE140

Amphiprotic Salt Hydrolysis

8. The pH of 0.1 M solution of (i) $NaHCO_3$, (ii) Na_2HPO_4 and (iii) NaH_2PO_4 is x, y and z, then the value of (y + z – x) is
(Roundoff your answer to the nearest integer)
Given that:

$H_2CO_3 + H_2O \rightleftharpoons H^{\oplus} + HCO_3^{\ominus}$; $K_1 = 4.2 \times 10^{-7}$ M

$HCO_3^{\ominus} \rightleftharpoons H^{\oplus} + CO_3^{2\ominus}$; $K_2 = 4.8 \times 10^{-11}$ M

$H_3PO_4 \rightleftharpoons H^{\oplus} + H_2PO_4^{\ominus}$; $K_1 = 7.5 \times 10^{-3}$ M

$H_2PO_4^{\ominus} \rightleftharpoons H^{\oplus} + HPO_4^{2\ominus}$; $K_2 = 6.2 \times 10^{-8}$ M

$HPO_4^{2\ominus} \rightleftharpoons H^{\oplus} + PO_4^{3\ominus}$; $K_3 = 1.0 \times 10^{-12}$ M

(log 4.2 = 0.62, log 4.8 = 0681, log 6.2 = 0.80, log 7.5 = 0.88)

CIE144

For More Join: @IITJEE_Advanced

9. The pH of 0.05M potassium hydrogen phthalate, $KHC_8H_4O_4$ is 'x', then the value of 100x is

$H_2C_8H_4O_4 + H_2O \rightleftharpoons H_3O^{\oplus} + HC_8H_4O_4^{\ominus}$; $pK_1 = 2.94$

$HC_8H_4O_4^{\ominus} + H_2O \rightleftharpoons H_3O^{\oplus} + C_8H_4O_4^{2\ominus}$; $pK_2 = 5.44$

CIE142

Solubility Equilibria Of Sparingly Soluble Salts, Solubility And Solubility Product Constant (K_{sp})

10. Calculate the solubility of AgCN in a buffer solution at pH = 3.0

Given : K_{sp} of AgCN = 8×10^{-10} , K_a of HCN = 5×10^{-10}. If the solubility of AgCN is $x \times 10^{-2}$ M, then x is

CIE028

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Electrolytes, Ionization Constant Of Water, Ionic Product Of Water (Kw), Ionization Of Acids And Bases

Rankers Academy JEE

11. 10 ml of 0.1M weak acid HA($k_a = 10^{-5}$) is mixed with 10 ml 0.2M HCl and 10 ml 0.1M NaOH. Find the value of $[A^-]$ in the resulting solution.

CIE234

Calculation Of pH Of Solution Of Acid-Base Mixtures Containing Weak Acids And Weak Bases

12. An aqueous solution contains 0.01 M RNH_2 ($K_b = 2 \times 10^{-6}$) & 10^{-4} M NaOH. The concentration of OH^- is nearly :

CIE228

Calculation Of pH Of Solution Containing Di- And Polybasic Acids And Di-And Polyacidic Bases

13. What are the concentrations of H^+, HSO_4^-;SO_4^{2-} and H_2SO_4 in a 0.20 M solution of sulphuric acid?

Given : $H_2SO_4 \longrightarrow H^+ + HSO_4^-$; strong

$HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$; $K_2 = 10^{-2}$ M

CIE224

14. Calculate the pH of a 0.1M solution of $H_2NCH_2CH_2NH_2$; ethylenediamine (en). Determine the en H_2^{2+} concentration in the solution. K_{b_1} and K_{b_2} values of ethylenediamine are 9×10^{-5} and 7.1×10^{-8} respectively.

CIE225

For More Join: @IITJEE_Advanced

15. Nicotine, $C_{10}H_{14}N_2$, has two basic nitrogen atoms and both can react with water to give a basic solution

$Nic\ (aq) + H_2O \rightleftharpoons NicH^+\ (aq) + OH^-\ (aq)$

$NicH^+\ (aq) + H_2O \rightleftharpoons NicH_2^{2+}\ (aq) + OH^-\ (aq)$

K_{b_1} is 8×10^{-7} and K_{b_2} is 10^{-10}. Calculate the approximate pH of a 0.20 M solution.

CIE226

Salt And Types Of Salts, Hydrolysis Of Salts And The pH Of Their Solutions, Monovalent Salts

16. Calculate the pH of 1.0×10^{-3} M sodium phenoxide, $NaOC_6H_5$. K_a for HOC_6H_5 is 0.6×10^{-10}.

CIE229

17. Calculate the pH of a 2.0 M solution of NH_4Cl. [K_b (NH_3) = 2×10^{-5}]

CIE145

18. 150 ml of 0.5 M HCN (Ka = 3.75×10^{-9}) was reacted with 1.5 M KOH for complete neutralisation. What will be molarity of HCN at equilibrium.

CIE235

Hydrolysis Of Polyvalent Salts

19. Calculate the OH^- concentration and the H_3PO_4 concentration of a solution prepared by dissolving 0.1 mol of $Na_3\ PO_4$ in sufficient water to make 1L of solution. $K_1 = 7.1 \times 10^{-3}$, $K_2 = 6.3 \times 10^{-8}$, $K_3 = 4.5 \times 10^{-13}$.

CIE230

Buffer Solution, Designing Of Buffer Solution, pH Of Buffer

Rankers Academy JEE

20. An ammonia-ammonium chloride buffer has a pH value of 9 with [NH_3] = 0.25. What will be the new pH if 500 ml 0.1 M KOH is added to 200 ml buffer solution ($K_b = 2 \times 10^{-5}$)
[log 2 = 0.3]

CIE232

Indicators

21. The indicator phenol red is half in the ionic form when pH is 7.2. If the ratio of the undissociated form to the ionic form is 1 : 5, find the pH of the solution. With the same pH for solution, if indicator is altered such that the ratio of undissociated form to dissociated form becomes 1 : 4, find the pH when 50 % of the new indicator is in ionic form. [log2 = 0.3]

CIE236

pH Curve

22. A weak base (50.0mL) was titrated with 0.1 M HCl. The pH of the solution after the addition of 10.0 mL and 25.0 mL were found to be 9.84 and 9.24, respectively. Calculate K_b of the base and pH at the equivalence point. [log2 = 0.3]

CIE233

Simultaneous Solubility, Solubility Of A_2B, AB_2 Type Of Salt And Solubility Of Complex Salts

23. Determine the [S^{2-}] in a saturated (0.1M) H_2S solution to which enough HCl has been added to produce a [H^+] of 2×10^{-4}. $K_1 = 10^{-7}$, $K_2 = 10^{-14}$.

CIE227

For More Join: @IITJEE_Advanced

24. How much AgBr could dissolve in 1.0 L of 0.40 M NH_3 ? Assume that $Ag(NH_3)_2^+$ is the only complex formed. [$K_f(Ag(NH_3)_2^+) = 1 \times 10^8$; K_{sp} (AgBr) = 5×10^{-13}] [$\sqrt{50} \simeq 7$]

CIE237

25. Calculate solubility of PbI_2 ($K_{sp} = 1.4 \times 10^{-8}$) in water at 25°, which is 90% dissociated.

$\left(\frac{1.4}{(0.81)(3.6)}\right)^{1/3} = 0.78$

CIE238

26. A recent investigation of the complexation of SCN^- with Fe^{3+} led to 130, 16, and 1.0 for K_1, K_2, and K_3, respectively. What is the overall formation constant of $Fe(SCN)_3$ from its component ions, and what is the dissociation constant of $Fe(SCN)_3$ into its simplest ions on the basis of these data ?

CIE239

27. Calculate the minimum mass of Na_2SO_4 needed to just start precipitation of $BaSO_4$ from 500 ml of 2×10^{-5}M - $BaCl_2$ solution. K_{sp} of $BaSO_4 = 8 \times 10^{-8}$.

CIE185

28. The pH of 0.1M $NaHCO_3$ is x, then the value of 10x is
($K_1 = 5 \times 10^{-7}$, $K_2 = 5 \times 10^{-11}$ for carbonic acids).

CIE141

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. If 1 mM solution of ethylamine produces pH = 9, then the ionization constant (K_b) of ethylamine is 10^{-x}. The value of x is _____ (nearest integer).
[The degree of ionization of ethylamine can be neglected with respect to unity.] **[JEE (Main) 2025]**

CIE320

2. A weak acid HA has degree of dissociation x. Which option gives the correct expression of pH = pK_a) ? **[JEE (Main) 2025]**

(1) log (1 + 2x) (2) $\log\left(\frac{1-x}{x}\right)$ (3) 0 (4) $\log\left(\frac{x}{1-x}\right)$

CIE321

3. For the equilibrium,
$2H_2O \rightleftharpoons H_3O^{\oplus} + OH^{\ominus}$, the value of $\Delta G°$ at 298 K is approximately:- **[JEE (Main) 2019]**

(1) –80 kJ mol^{-1} (2) –100 kJ mol^{-1} (3) 100 kJ mol^{-1} (4) 80 kJ mol^{-1}

CIE319

4. pH of water is 7 at 25°C. If water is heated to 80°C, it's pH will : **[JEE (Main) 2025]**
(1) Decrease
(2) Remains the same
(3) H^+ concentration increases, OH^- concentration decreases
(4) Increase

CIE322

Pankers Academy JEE

5. When the hydrogen ion concentration [H^+] changes by a factor of 1000, the value of pH of the solution _____. **[JEE (Main) 2023]**

(1) increases by 1000 units (2) decreases by 3 units
(3) decreases by 2 units (4) increases by 2 units

CIE248

6. Millimoles of calcium hydroxide required to produce 100 mL of the aqueous solution of pH 12 is $x \times 10^{-1}$. The value of x is _____ (Nearest integer).
Assume complete dissociation. **[JEE (Main) 2023]**

CIE249

7. 600 mL of 0.01M HCl is mixed with 400 mL of 0.01 M H_2SO_4. The pH of the mixture is_____ $\times 10^{-2}$.
(Nearest integer)
[Given log 2 = 0.30, log 3 = 0.48, log 5 = 0.69, log 7 = 0.84, log 11 = 1.04] **[JEE (Main) 2023]**

CIE250

8. An analyst wants to convert. 1L HCl of pH = l to a solution of HCl of pH 2. The volume of water needed to do this dilution is _____ mL. (Nearest Integer) **[JEE (Main) 2023]**

CIE255

9. K_a for butyric acid (C_3H_7COOH) is 2×10^{-5}. The pH of 0.2 M solution of butyric acid is __ $\times 10^{-1}$. (Nearest integer) [Given log 2 = 0.30] **[JEE (Main) 2022]**

CIE244

10. K_a for CH_3COOH is 1.8×10^{-5} and K_b for NH_4OH is 1.8×10^{-5}. The pH of ammonium acetate solution will be **[JEE (Main) 2024]**

CIE311

For More Join: @IITJEE_Advanced

11. Given below are two statements :

Statement (I) : Aqueous solution of ammonium carbonate is basic.

Statement (II) : Acidic/basic nature of salt solution of a salt of weak acid and weak base depends on K_a and K_b value of acid and the base forming it.

In the light of the above statements, choose the most appropriate answer from the options given below : **[JEE (Main) 2024]**

(1) Both Statement I and Statement II are correct

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are incorrect

(4) Statement I is incorrect but Statement II is correct

CIE312

12. 20 mL of 0.1 M NaOH is added to 50 mL of 0.1 M acetic acid solution. The pH of the resulting solution is _____ $\times 10^{-2}$ (Nearest integer)

Given : pKa (CH_3COOH) = 4.76, log 2 = 0.30, log 3 = 0.48 **[JEE (Main) 2023]**

CIE257

13. The pH of an aqueous solution containing 1M benzoic acid (pK_a = 4.20) and 1M sodium benzoate is 4.5. The volume of benzoic acid solution in 300 mL of this buffer solution is _____ mL.

[JEE (Main) 2024]

CIE313

14. Given below are two statements :

Statement (I) : A Buffer solution is the mixture of a salt and an acid or a base mixed in any particular quantities.

Statement (II) : Blood is naturally occurring buffer solution whose pH is maintained by $H_2CO_3 / HCO_3^{\ominus}$ concentrations.

In the light of the above statements, choose the **correct** answer from the options given below.

[JEE (Main) 2024]

(1) **Statement I** is false but **Statement II** is true

(2) Both **Statement I** and **Statement II** is true

(3) Both **Statement I** and **Statement II** is false

(4) **Statement I** is true but **Statement II** is false

CIE314

15. A litre of buffer solution contains 0.1 mole of each of NH_3 and NH_4Cl. On the addition of 0.02 mole of HCl by dissolving gaseous HCl, the pH of the solution is found to be _____$\times 10^{-3}$ (Nearest integer)

[**Given** : $pK_b(NH_3)$ = 4.745; log 2 = 0.301; log 3 = 0.477; T = 298 K] **[JEE (Main) 2023]**

CIE247

16. Class XII students were asked to prepare one litre of buffer solution of pH 8.26 by their chemistry teacher. The amount of ammonium chloride to be dissolved by the student in 0.2 M ammonia solution to make one litre of the buffer is (Given pK_b (NH_3) = 4.74; Molar mass of NH_3 = 17 g mol^{-1}; Molar mass of NH_4Cl = 53.5 g mol^{-1}) **[JEE (Main) 2022]**

(1) 53.5 g (2) 72.3 g (3) 107.0 g (4) 126.0 g

CIE240

For More Join: @IITJEE_Advanced

17. Given below are two statements :

Statement (I) : Potassium hydrogen phthalate is a primary standard for standardisation of sodium hydroxide solution.

Statement (II) : In this titration phenolphthalein can be used as indicator.

In the light of the above statements, choose the **most appropriate** answer from the options given below: **[JEE (Main) 2024]**

(1) Both Statement I and Statement II are correct

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are incorrect

CIE315

18. Given below are two statements :

Statement I :- Methyl orange is a weak acid.

Statement II :- The benzenoid form of methyl orange is more intense/deeply coloured than the quinonoid form.

In the light of the above statement, choose the most appropriate answer from the options given below : **[JEE (Main) 2023]**

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but statement II is correct.

(3) Both Statement I and Statement II are incorrect.

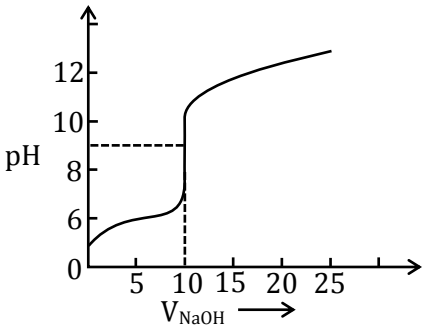
(4) Both statement I and Statement II are correct.

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CIE253

19. The titration curve of weak acid vs. strong base with phenolphthalein as indictor) is shown below. The $K_{phenolphthalein} = 4 \times 10^{-10}$.

Given: log2 = 0.3



The number of following statements which is/are correct about phenolphthalein is_____

(1) It can be used as an indicator for the titration of weak acid with weak base.

(2) It begins to change colour at pH = 8.4

(3) It is a weak organic base

(4) It is colourless in acidic medium **[JEE (Main) 2023]**

CIE252

For More Join: @IITJEE_Advanced

20. Which of the following statement(s) is/are correct?

(A) The pH of 1×10^{-8} M HCl solution is 8.

(B) The conjugate base $H_2PO_4^-$ is HPO_4^{2-}.

(C) K_w increases with increase in temperature.

(D) When a solution of weak monoprotic acid is titrated against a strong base at half neutralisation point, $pH = \frac{1}{2}pK_a$

Choose the correct answer from the option given below. **[JEE (Main) 2023]**

(1) (B), (C), (D) (2) (A), (D) (3) (A), (B), (C) (4) (B), (C)

CIE258

21. Incorrect statement for the use of indicators in acid-base titration is : **[JEE (Main) 2023]**

(1) Methyl orange may be used for a weak acid vs weak base titration.

(2) Methyl orange is a suitable indicator for a strong acid vs weak base titration

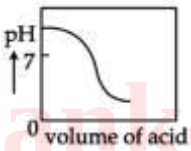
(3) Phenolphthalein is a suitable indicator for a weak acid vs strong base titration

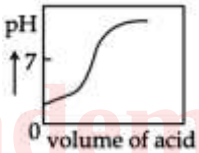
(4) Phenolphthalein may be used for a strong acid vs strong base titration.

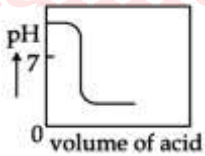
CIE259

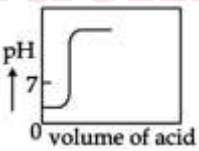
22. The Plot of pH-metric titration of weak base NH_4OH vs strong acid HCl looks like:

[JEE (Main) 2022]

(1) 

(2) 

(3) 

(4) 

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CIE241

23. 50 mL of 0.1 M CH_3COOH is being titrated against 0.1 M NaOH. When 25 mL of NaOH has been added, the pH of the solution will be _____ $\times 10^{-2}$. (Nearest integer) **[JEE (Main) 2022]**

(Given : pK_a (CH_3COOH) = 4.76)

log 2 = 0.30

log 3 = 0.48

log 5 = 0.69

log 7 = 0.84

log 11 = 1.04

CIE243

24. Ksp for $Cr(OH)_3$ is 1.6×10^{-30}. What is the molar solubility of this salt in water? **[JEE (Main) 2025]**

(1) $\sqrt[4]{\frac{1.6\times10^{-30}}{27}}$ (2) $\frac{1.8\times10^{-30}}{27}$ (3) $\sqrt[5]{1.8\times10^{-30}}$ (4) $\sqrt[2]{1.6\times10^{-30}}$

CIE323

For More Join: @IITJEE_Advanced

25. The molar solubility(s) of zirconium phosphate with molecular formula $(Zr^{4+})_3\left(PO_4^{3-}\right)_4$ is given by relation : **[JEE (Main) 2025]**

(1) $\left(\frac{K_{sp}}{6912}\right)^{\frac{1}{7}}$ (2) $\left(\frac{K_{sp}}{5348}\right)^{\frac{1}{6}}$ (3) $\left(\frac{K_{sp}}{8435}\right)^{\frac{1}{7}}$ (4) $\left(\frac{K_{sp}}{9612}\right)^{\frac{1}{3}}$

CIE324

26. At 298 K, the solubility of silver chloride in water is $1.434\times10^{-3}\ g\ L^{-1}$. The value of $-\log K_{sp}$ for silver chloride is _____.

(Given mass of Ag is 107.9 $g\ mol^{-1}$ and mass of Cl is $35.5\ g\ mol^{-1}$) **[JEE (Main) 2023]**

CIE251

27. Which of the following happens when NH_4OH is added gradually to the solution containing 1M A^{2+} and 1M B^{3+} ions ?

Given : $K_{sp}[A(OH)_2] = 9 \times 10^{-10}$ and

$K_{sp}[B(OH)_3] = 27 \times 10^{-18}$ at 298 K. **[JEE (Main) 2025]**

(1) $B(OH)_3$ will precipitate before $A(OH)_2$

(2) $A(OH)_2$ and $B(OH)_3$ will precipitate together

(3) $A(OH)_2$ will precipitate before $B(OH)_3$

(4) Both $A(OH)_2$ and $B(OH)_3$ do not show precipitation with NH_4OH

CIE325

28. The pH at which $Mg(OH)_2$ [$K_{sp} = 1 \times 10^{-11}$] begins to precipitate from a solution containing 0.10 M Mg^{2+} ions is _____ **[JEE (Main) 2024]**

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CIE316

29. Solubility of calcium phosphate (molecular mass, M) in water is W_g per 100 mL at 25° C. Its solubility product at 25°C will be approximately. **[JEE (Main) 2024]**

(1) $10^7\left(\frac{W}{M}\right)^3$ (2) $10^7\left(\frac{W}{M}\right)^5$ (3) $10^3\left(\frac{W}{M}\right)^5$ (4) $10^5\left(\frac{W}{M}\right)^5$

CIE317

30. For a sparingly soluble salt AB_2, the equilibrium concentrations of A^{2+} ions and B^- ions are 1.2×10^{-4} M and 0.24×10^{-3} M, respectively. The solubility product of AB_2 is : **[JEE (Main) 2024]**

(1) 0.069×10^{-12} (2) 6.91×10^{-12} (3) 0.276×10^{-12} (4) 27.65×10^{-12}

CIE318

31. 25.0 mL of 0.050 M $Ba(NO_3)_2$ is mixed with 25.0 mL of 0.020 M NaF. K_{sp} of BaF_2 is 0.5×10^{-6} at 298 K. The ratio of $\left[Ba^{2+}\right]\left[F^-\right]^2$ and K_{sp} is _____ (Nearest integer) **[JEE (Main) 2023]**

CIE256

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Dilution process of different aqueous solutions; with water, are given in LIST-I. The effects of dilution of the solutions on $[H^{\oplus}]$ are given in LIST-II.

(Note : Degree of dissociation (α) of weak acid and weak base is << 1; degree of hydrolysis of salt <<1; $[H^{\oplus}]$ represents the concentration of $H^{\oplus}$ ions)

LIST-I		List-II	
P.	(10 mL of 0.1 M NaOH + 20 mL of 0.1 M acetic acid) diluted to 60 mL	1	the value of $[H^{\oplus}]$ does not change on dilution
Q.	(20 mL of 0.1 M NaOH + 20 mL of 0.1 M acetic acid) diluted to 80 mL	2	the value of $[H^{\oplus}]$ change to half of its initial value on dilution
R.	(20 mL of 0.1 M HCl + 20 mL of 0.1 M ammonia solution) diluted to 80 mL	3	the value of $[H^{\oplus}]$ changes to two times of its initial value on dilution
S.	10 mL saturated solution of $Ni(OH)_2$ in equilibrium with excess solid $Ni(OH)_2$ is diluted to 20 mL (solid $Ni(OH)_2$ is still present after dilution).	4	the value of $[H^{\oplus}]$ changes to $\frac{1}{\sqrt{2}}$ times of its initial value on dilution
		5	the value of $[H^{\oplus}]$ changes to $\sqrt{2}$ times of its initial value on dilution

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Match each process given in LIST-I with one or more effect(s) in LIST-II. The correct option is

[JEE (Advanced) 2018]

(A) P → 4; Q → 2; R → 3; S → 1 (B) P → 4; Q → 3; R → 2; S → 3

(C) P → 1; Q → 4; R → 5; S → 3 (D) P → 1; Q → 5; R → 4; S → 1

CIE093

2. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA, 1M) is $1/100^{th}$ of that of a strong acid (HX, 1M), at 25°C. The K_a of HA is **[JEE (Advanced) 2013]**

(A) 1×10^{-4} (B) 1×10^{-5} (C) 1×10^{-6} (D) 1×10^{-3}

CIE088

3. A solution is prepared by mixing 0.01 mol each of H_2CO_3, $NaHCO_3$, Na_2CO_3, and NaOH in 100 mL of water. pH of the resulting solution is _____.

[Given : pK_{a1} and pK_{a2} of H_2CO_3 are 6.37 and 10.32, respectively ; log 2 = 0.30]

[JEE (Advanced) 2022]

CIE097

4. On decreasing the *p*H from 7 to 2, the solubility of a sparingly soluble salt (MX) of a weak acid (HX) increased from 10^{-4} mol L^{-1} to 10^{-3} mol L^{-1}. The pK_a of HX is:

(A) 3 (B) 4 (C) 5 (D) 2

[JEE (Advanced) 2023]

CIE260

For More Join: @IITJEE_Advanced

5. Concentration of H_2SO_4 and Na_2SO_4 in a solution is 1 M and 1.8×10^{-2} M, respectively. Molar solubility of $PbSO_4$ in the same solution is $X \times 10^{-Y}$ M (expressed in scientific notation). The value of Y is _____.

[Given: Solubility and Solubility Product of $PbSO_4$ (K_{sp}) = 1.6×10^{-8}. For H_2SO_4, K_{a_1} is very large and $K_{a_2} = 1.2 \times 10^{-2}$]

[JEE (Advanced) 2022]

CIE098

6. An acidified solution of 0.05 M $Zn^{2\oplus}$ is saturated with 0.1 M H_2S. What is the minimum molar concentration (M) of $H^{\oplus}$ required to prevent the precipitation of ZnS ?

Use K_{sp} (ZnS) = 1.25×10^{-22} and overall dissociation constant of H_2S, $K_{NET} = K_1K_2 = 1 \times 10^{-21}$

[JEE (Advanced) 2020]

CIE095

7. The solubility of a salt of weak acid(AB) at pH=3 is $Y \times 10^{-3}$ mol L^{-1}. The value of Y is___.

(Given that the value of Solubility and Solubility Product of AB (K_{sp}) $= 2 \times 10^{-10}$ and the value of ionization constant of HB(K_a) $= 1 \times 10^{-8}$)

[JEE (Advanced) 2018]

CIE092

8. The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K. The solubility (in mol/L) of Ag_2CrO_4 in a 0.1 M $AgNO_3$ solution is

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[JEE (Advanced) 2013]

(A) 1.1×10^{-11} (B) 1.1×10^{-10} (C) 1.1×10^{-12} (D) 1.1×10^{-9}

CIE089

9. In 1 L saturated solution of AgCl [K_{sp}(AgCl) $= 1.6 \times 10^{-10}$], 0.1 mol of CuCl [K_{sp}(CuCl) $= 1.0 \times 10^{-6}$] is added. The resultant concentration of $Ag^{\oplus}$ in the solution is 1.6×10^{-x}. The value of 'x' is.

[JEE (Advanced) 2011]

CIE087

10. 5.00 mL of 0.10 M oxalic acid solution taken in a conical flask is titrated against NaOH from a burette using phenolphthalein indicator. The volume of NaOH required for the appearance of permanent faint pink color is tabulated below for five experiments. What is the concentration, in molarity, of the NaOH solution?

[JEE (Advanced) 2020]

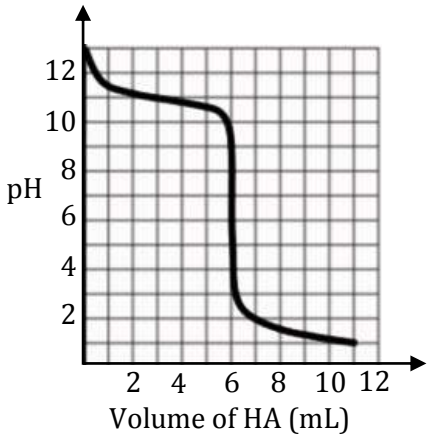
Exp. No.	Vol. of NaOH (mL)
1	12.5
2	10.5
3	9.0
4	9.0
5	9.0

CIE094

For More Join: @IITJEE_Advanced

11. A solution of 0.1 M weak base (B) is titrated with 0.1 M of a strong acid (HA). The variation of pH of the solution with the volume of HA added is shown in the figure below. What is the pK_b of the base? The neutralization reaction is given by $B + HA \rightarrow BH^{\oplus} + A^{\ominus}$.

[JEE (Advanced) 2020]



CIE096

Paragraph For Questions 12 and 13

When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increases of 5.7°C was measured for the beaker and its content. **(Expt-1).** Because the enthalpy of neutralisation of a strong acid with a strong base is a constant ($-57.0\ kJmol^{-1}$), this experiment could be used to measure the calorimeter constant. In a second experiment **(Expt-2)**, 100 mL of 2.0 M acetic acid ($K_a = 2.0 \times 10^{-5}$) was mixed with 100 mL of 1.0M NaOH (under identical conditions to **(Expt-1)**) where a temperature rise of 5.6°C was measured. (Consider heat capacity of all solutions as $4.2\ Jg^{-1}K^{-1}$ and density of all solutions as $1.0\ g\ mL^{-1}$)

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12. Enthalpy of dissociation (in $kJ\ mol^{-1}$) of acetic acid obtained from the **Expt-2** is

[JEE (Advanced) 2015]

(A) 1.0 (B) 10.0 (C) 24.5 (D) 51.4

CIE090

13. The pH of the solution after **Expt-2**

[JEE (Advanced) 2015]

(A) 2.8 (B) 4.7 (C) 5.0 (D) 7.0

CIE091

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	D	A	D	C	B	C	B	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	D	D	D	C	B	A	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	C	B	B	D	B	A	B	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	B	C	C	C	A	A	D	A	D
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	B	A	C	B	C	C	D	B	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	A	A	B	D	A	C	C	A	D	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	B	B	C	C	C	A	D	A	C
Que.	71	72	73	74	75	76				
Ans.	B	D	B	B	A	B				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABC	ABCD	BC	AB	ABCD	ACD	ABD	ABD	AC	ABCD
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	BC	C	B	D	A	C	C	C	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	C	C	A	B	B	B	B	A
Que.	31									
Ans.	C									

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EXERCISE (S)

1. 6 **2.** 3 **3.** 7 **4.** 5

5. 6 **6.** 25 **7.** 48 **8.** 6

9. 419 **10.** 4 **11.** 10^{-5} **12.** 2×10^{-4}

13. 0.209 M, 0.191 M, 9.13×10^{-3} M, 0

14. pH = 11.48 , $[enH_2^{2+}] = 7.1 \times 10^{-8}$ M

15. 10.6 **16.** pH = 10.52 **17.** 4.5

18. 10^{-3}

19. $[OH^-] = 3.73 \times 10^{-2}$M, $[H_3PO_4] = 6 \times 10^{-18}$M **20.** 9.6

21. pH = 7.9 , pH = 7.3 **22.** $K_b \simeq 1.73 \times 10^{-5}$, 5.27

23. $[S^{2-}] = 2.5 \times 10^{-15}$M **24.** 2.8×10^{-3} mole **25.** 1.6×10^{-3}

26. $K_d = 1/K_f = 4.8 \times 10^{-4}$ **27.** (0.284) **28.** 83

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	7	4	4	1	2	5	186	9000	27	7
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	458	100	1	9079	3	1	3	2	4
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	476	1	1	10	1	09	2	2
Que.	31									
Ans.	5									

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	10.02	B	6	0.20	4.47	B	7	0.11
Que.	11	12	13							
Ans.	2.30 to 3.00	A	B							

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